

ZS Associates SQL Practice Worksheet

Employee(emp_id, name, dept_id, salary, hire_date)

emp_id	name	dept_id	salary	hire_date
1	Alice	10	70000	2023-01-10
2	Bob	10	60000	2022-05-15
3	Carol	20	90000	2021-08-01
4	David	30	50000	2024-02-11
5	Eva	20	95000	2020-09-30
6	Frank	30	55000	2023-07-18

Department(dept_id, dept_name)

dept_id	dept_name
10	HR
20	IT
30	Sales

Orders(order_id, customer_id, amount, order_date)

order_id	customer_id	amount	order_date
101	1	500	2026-01-10
102	2	700	2026-01-11
103	1	300	2026-02-10
104	3	900	2026-02-15
105	4	100	2026-03-01

Practice Questions

1. Display all employees.
2. Employees with salary > 60000.
3. Employees hired after 2023-01-01.
4. Count employees in each department.
5. Departments having more than 1 employee.
6. Average salary by department.
7. Highest salary in each department.
8. Employees earning above company average.
9. Second highest salary.

10. Top 3 highest salaries.
11. Join Employee and Department to show employee and department name.
12. Left join to show all departments with employee count.
13. Find employees in IT department.
14. Rank employees by salary within department.
15. Dense rank salaries overall.
16. Running total of salaries ordered by emp_id.
17. Find duplicate salaries.
18. Total order amount by customer.
19. Customer with highest purchase.
20. Customers with more than one order.
21. Monthly sales.
22. Customers with no orders (assume Customer table).
23. Use CASE to classify salary (>80000 High, >60000 Medium, else Low).
24. Employees whose salary is above department average.
25. Write a CTE to find top earner per department.
26. Use ROW_NUMBER to return highest paid employee in each department.
27. Find employees hired in the last 2 years.
28. Difference between max and min salary in each department.
29. Update salary of Sales employees by 10% (write query only).
30. Delete duplicate rows keeping lowest emp_id (write query pattern).